Christopher Sparrow

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Available for Spring/Summer 2027 Co-op

EDUCATION

Northeastern University – Boston, MA **Expected May 2029**
Bachelor of Science in Electrical Engineering **GPA: 3.84**
Honors: Dean's List
Coursework: Embedded Design, Circuits and Signals, Differential Equations and Linear Algebra, Physics II
Affiliations: IEEE Student Member, MTT-S (Microwave Theory), AP-S (Antennas and Propagation), AESS (Aerospace)

TECHNICAL SKILLS

Languages: C/C++, MicroPython, MATLAB, Python
Hardware & Protocols: FPGA, ARM Cortex-M, Raspberry Pi Pico, Arduino, PWM, I2C, UART, SPI, SWD, JTAG
Design Tools: KiCad, LTSpice, PlatformIO, Quartus, Git, Linux, SolidWorks, AutoCAD, 3D Printing, Laser Cutting
Lab Equipment: Oscilloscope, Digital Multimeter, DC Power Supply, Function Generator

PROJECTS

Ground Station Telemetry Receiver (solo) – AerospaceNU *September 2026 – Present*
- Build a ground station that receives telemetry from team rocket avionics and relays decoded packets over Bluetooth to a laptop or phone for live tracking during flight
- Select the radio front end and antenna, lay out the board in KiCad, and write receive firmware

Custom GPS / Radio Avionics Printed Circuit Board (solo) – AerospaceNU *January 2026 – Present*
- Designed a custom printed circuit board in KiCad that integrates GPS and radio modules for rocket avionics
- Develop embedded drivers so the board can parse GPS position fixes and drive the radio without a host computer
- Order components, hand solder the assembly, and bench test the board for signal integrity and power draw
- Target a flight-ready board that logs position and transmits telemetry to a ground station through launch and recovery

Automatic Toothpaste Dispenser – Northeastern University *January 2026 – April 2026*
Electronics Lead, Team of 3
- Led electronics for an assistive dispenser that delivers dentist-recommended dose sizes for users with limited hand dexterity
- Programmed a Raspberry Pi Pico in MicroPython to drive a direct current gearmotor through a dual-channel motor driver, using pulse-width modulation and timed runs for adult and child dose settings
- Implemented 25 ms button debouncing and double-click detection so two inputs control dose size and motor direction

Avionics Flight Computer Firmware – AerospaceNU *September 2025 – December 2025*
- Worked alongside mentors to engineer embedded I2C drivers that convert raw MS5607 barometer and ICM-20602 registers into SI units and altitude
- Applied low-pass filtering to altitude and velocity data, reducing noise and eliminating false state triggers
- Programmed 50 Hz telemetry logging to S25FL512 SPI flash and state detection logic for liftoff, apogee, and landing

ACTIVITIES & LEADERSHIP

Northeastern University Launch Initiative (AerospaceNU) *Fall 2025 – Present*
Avionics Engineer
- Build the flight computers, telemetry, and control electronics that every subteam depends on
- Support projects such as automated air brakes and flight systems for a carbon fiber supersonic rocket

Northeastern Rugby Team (Division 1AA) *Fall 2025 – Present*
First Team Player
- Coach new forwards through scrums, lineouts, and pod plays, including team-specific calls
- Compete on a squad that went 6-0 in the 2025 Liberty Rugby Conference to earn the top playoff seed

INTERESTS

Diving: Freediving, underwater filmmaking, spearfishing, Advanced Open Water and Enriched Air Nitrox scuba certified
Other: Solo travel, chess, reading historical nonfiction, Texas Hold'em poker